

Towards Enriching Web Semantic Annotations for Automated Fact-Checking: **Schema.org-Guided Knowledge Graph Construction from News Articles with Large Language Models**

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M2 Thesis Internship

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Outline



- 01 **Motivation and objective**
- 02 **Background: key concepts**
- 03 **Related work and where this work sits**
- 04 **Methodology: the extraction pipeline**
- 05 **Experiments and results**
- 06 **Conclusion and perspectives**

Fact-checking needs facts the Web keeps in plain text

1 Automated fact-checking verifies claims against structured data

Knowledge graphs such as Wikidata and DBpedia give clean, machine-readable facts.

2 But knowledge graphs are incomplete, so the trustworthy Web must be mined

News sites hold recent facts as **raw text**, and curated KGs lag the news by weeks.

3 The bridge already exists: pages embed schema.org annotations

JSON-LD inside the HTML gives typed triples in a **shared web vocabulary**.

> 50% of crawled pages already carry schema.org triples

but the markup is **sparse** and describes the page, not the article's facts

LLM extraction can supply the missing content-grounded facts.

The Gap: Example

What the page publishes

```
{
  "@type": "NewsArticle",
  "headline": "France records excess
               deaths at peak of ...",
  "datePublished": "2026-07-03",
  "author": {
    "@type": "Person", ...
  },
  "publisher": {
    "@type": "Organization",
  }
}
```

What the page says

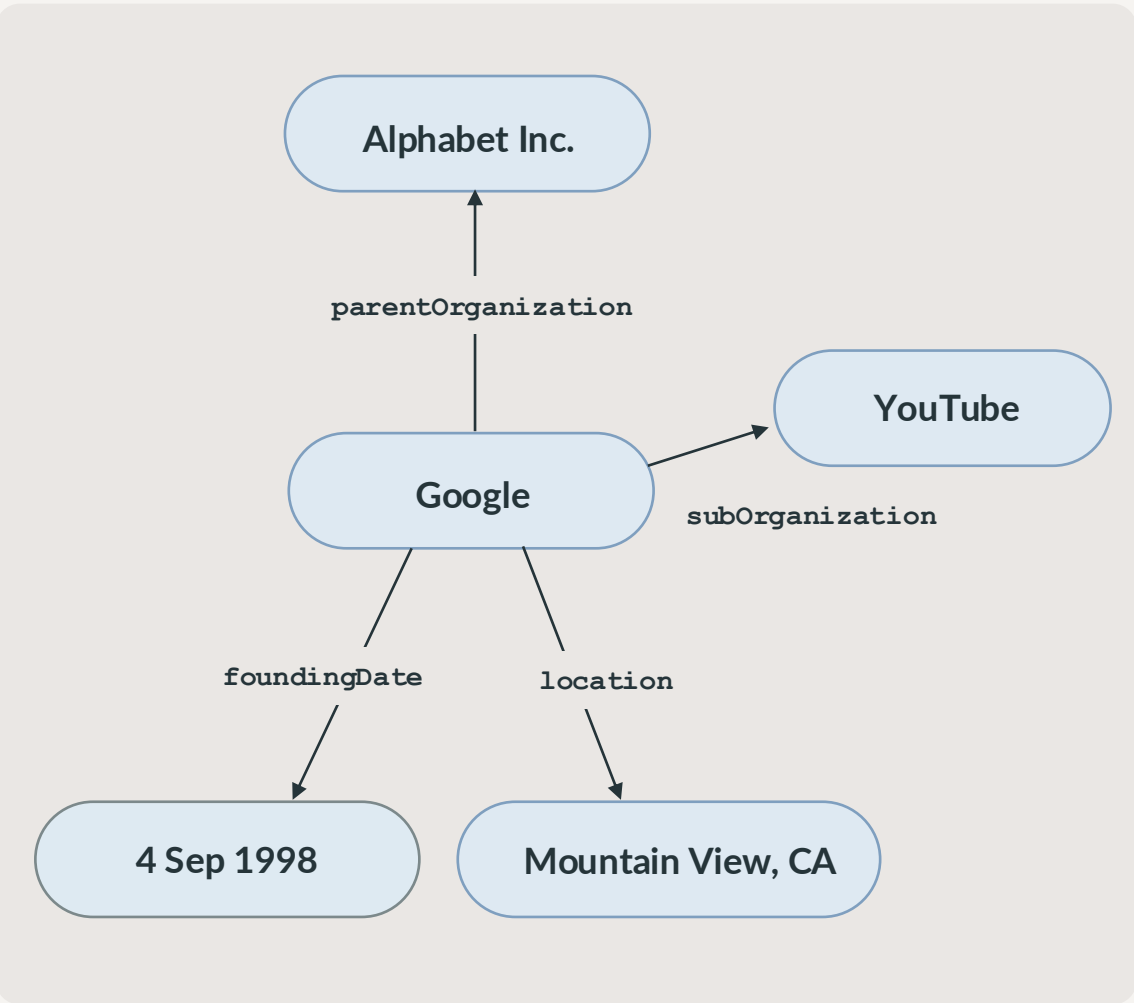
- A record-breaking heatwave affected Europe in June 2026
- France recorded **2,025 excess deaths** between 22 and 28 June
- That is a **29% rise** on the preceding week
- Belgium recorded **1,222 excess deaths**, 39% above normal
- The health minister **stated** there was a "clear increase" in deaths among those over 45

None of these facts is in the markup.
The markup describes the publication; the facts are in the text.

Background: Fundamental Concepts

Knowledge Graph

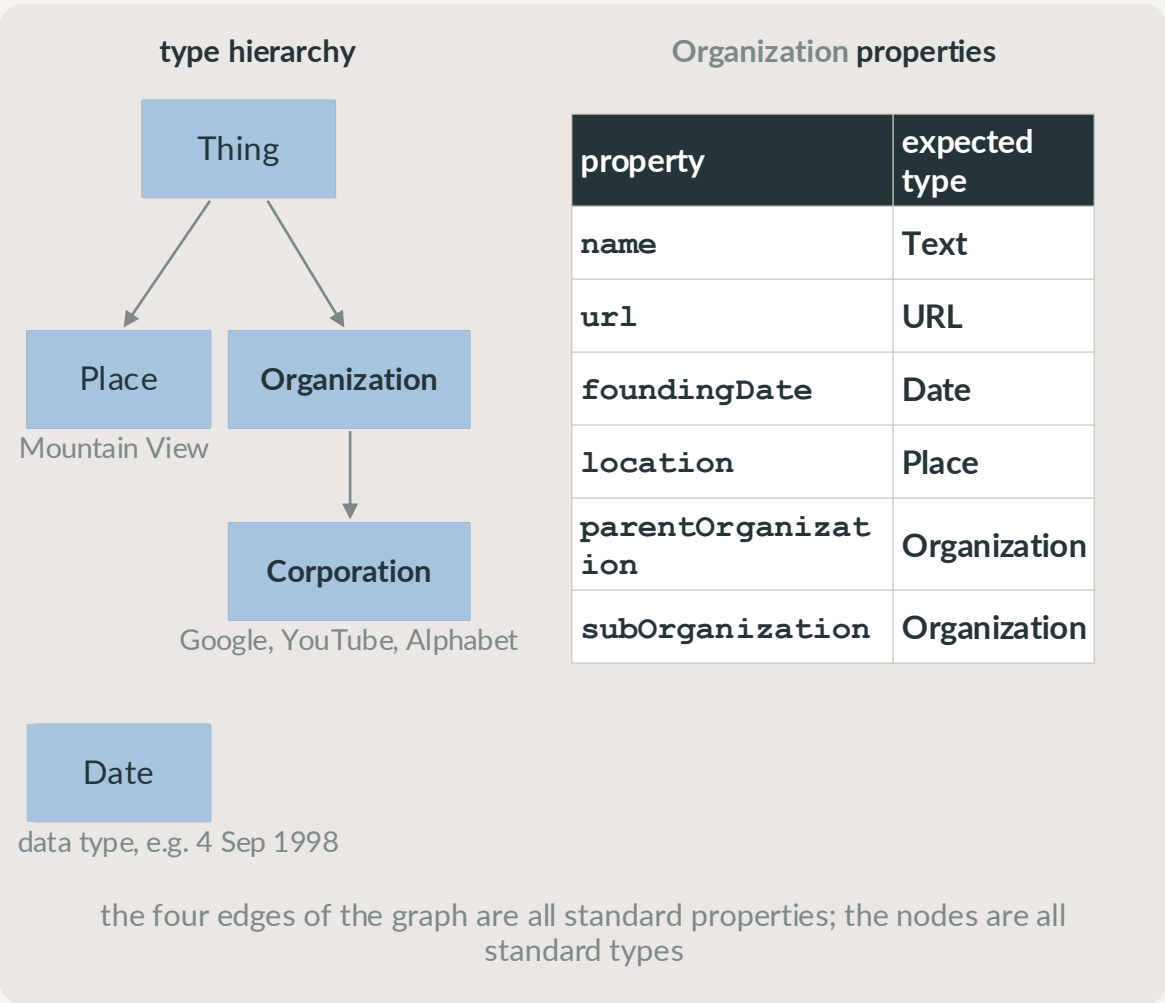
Facts as **triples** <subject, predicate, object>. Nodes are entities or values; edges are typed relations.



one triple: <Google, subOrganization, YouTube>

schema.org

A **vocabulary** of ~900 types in a hierarchy, ~1,500 properties with expected types. Subtypes inherit.



Semantic annotation

The same graph **embedded in a web page** as JSON-LD, in the schema.org vocabulary.

```
<script type="application/ld+json"> in the page's HTML
{
  "@context": "https://schema.org",
  "@type": "NewsArticle",
  "headline": "Google buys YouTube for $1.65B",
  "datePublished": "2006-10-09",
  "about": {
    "@type": "Corporation",
    "name": "Google",
    "foundingDate": "1998-09-04",
    "location": { "@type": "Place",
      "name": "Mountain View, CA" },
    "parentOrganization": {
      "@type": "Corporation",
      "name": "Alphabet Inc." },
    "subOrganization": {
      "@type": "Corporation",
      "name": "YouTube" }
  }
}
```

This thesis extracts the graph from the text, in the schema.org vocabulary, so it can be published as annotation.

OBJECTIVE

From a news article, extract a **schema.org-guided** and **grounded** knowledge graph using LLMs, where the vocabulary grows only for genuine gaps.

01 schema.org-guided

Every extracted type and relation is canonicalised against schema.org.

02 minimally extended vocab

A new term is minted only on a genuine no-match.

03 grounded

A fact carries the sentence that supports it from text.

Canonicalisation: aligning a free-form label to a standard vocabulary term.

Minting: creating a defined new term when nothing fits.

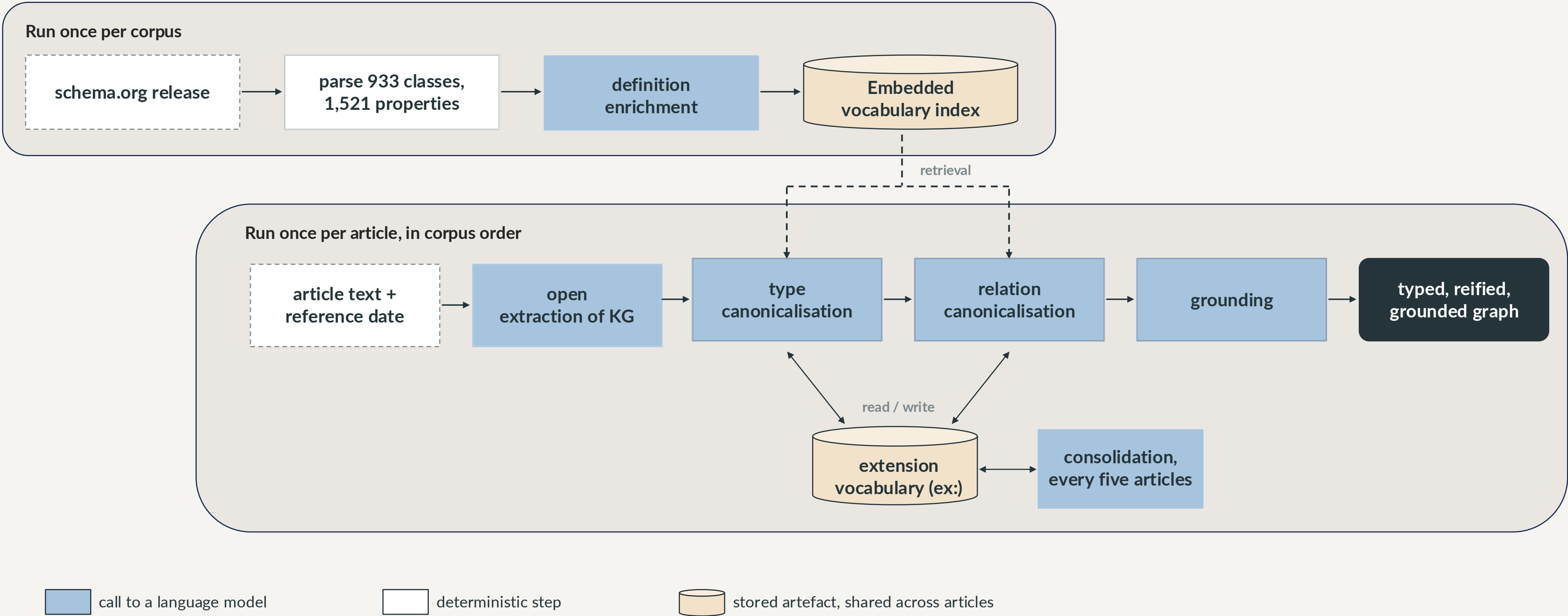
Related work: Where this work sits



System	Target vocabulary	Vocab in prompt	Input unit	Vocabulary growth	Fact shape
EDC (Zhang & Soh, 2024)	Wikidata / DBpedia relations	None	sentence	adds unmatched terms (self-canonicalisation)	binary
Text2KGBench (2023)	given small ontology	Full	sentence	fixed	binary
ODKE+ (2025)	KG ontology, 195 predicates	Partial	document	fixed	binary
KGGen (2025)	none, schema-free	-	document	–	binary
AutoSchemaKG (2025)	induced from corpus	-	chunked document	Induced from data	binary
LLM4Schema.org (2025)	schema.org	Partial	page text, chunked	fixed	page-level markup
This work	schema.org U ex:	None	full article	mint on logged no-match, then consolidate	Events, Claims + binary

No prior system combines these five choices

Pipeline Overview



Stages 2-3 • Open Extraction



Output Example



Evidence

EV1 event “Deaths rose 29% in the last week of June compared to the week before ...”
occurred in “France has said it recorded an increase of more than 2,000 deaths ...”

Definitions (one per free-form label)

ExcessMortality: deaths above the number expected for a period, attributed to a cause
occurred in: an association between an event and the place where it took place

LLM Call for Extraction, asks for:

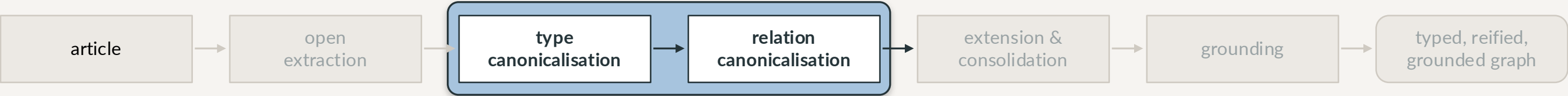
- **Entities** with a free-form type
- **Reified fact nodes** for events and claims
- **Directed relations**
- **A definition for every label**
- Evidence as **verbatim sentences**
- **Resolved temporal references** (wrt publication date)



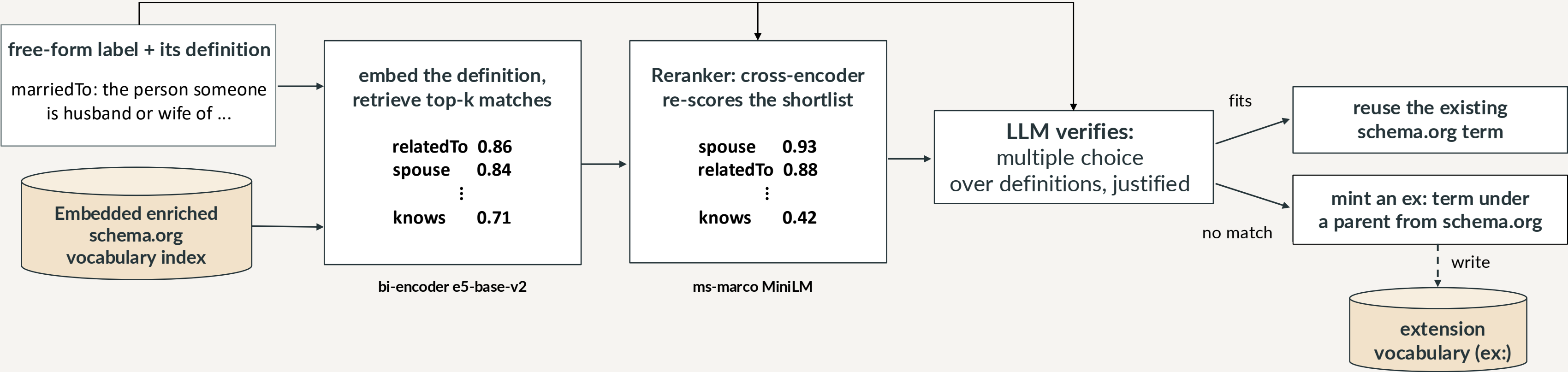
Deterministic Consistency Pass

Check duplicates & undefined endpoints

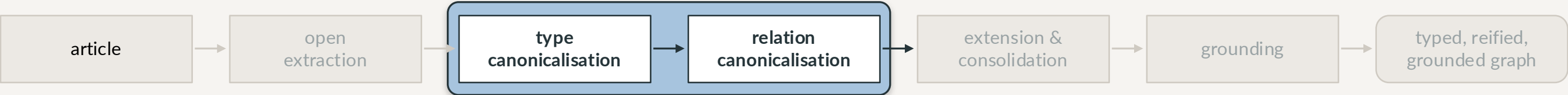
Stages 4-5 • Canonicalisation: Matching Terms to Schema.org



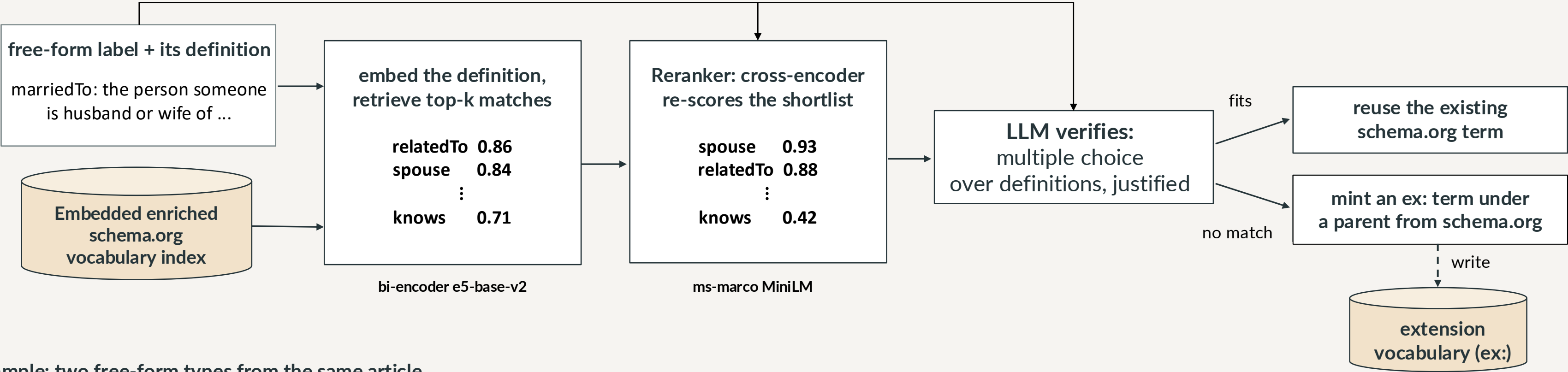
The Canonicalize step of EDC, plus reranking, with the accept-or-mint decision made explicitly rather than by a similarity threshold.



Stages 4-5 • Canonicalisation: Matching Terms to Schema.org



The Canonicalize step of EDC, plus reranking, with the accept-or-mint decision made explicitly rather than by a similarity threshold.



Example: two free-form types from the same article

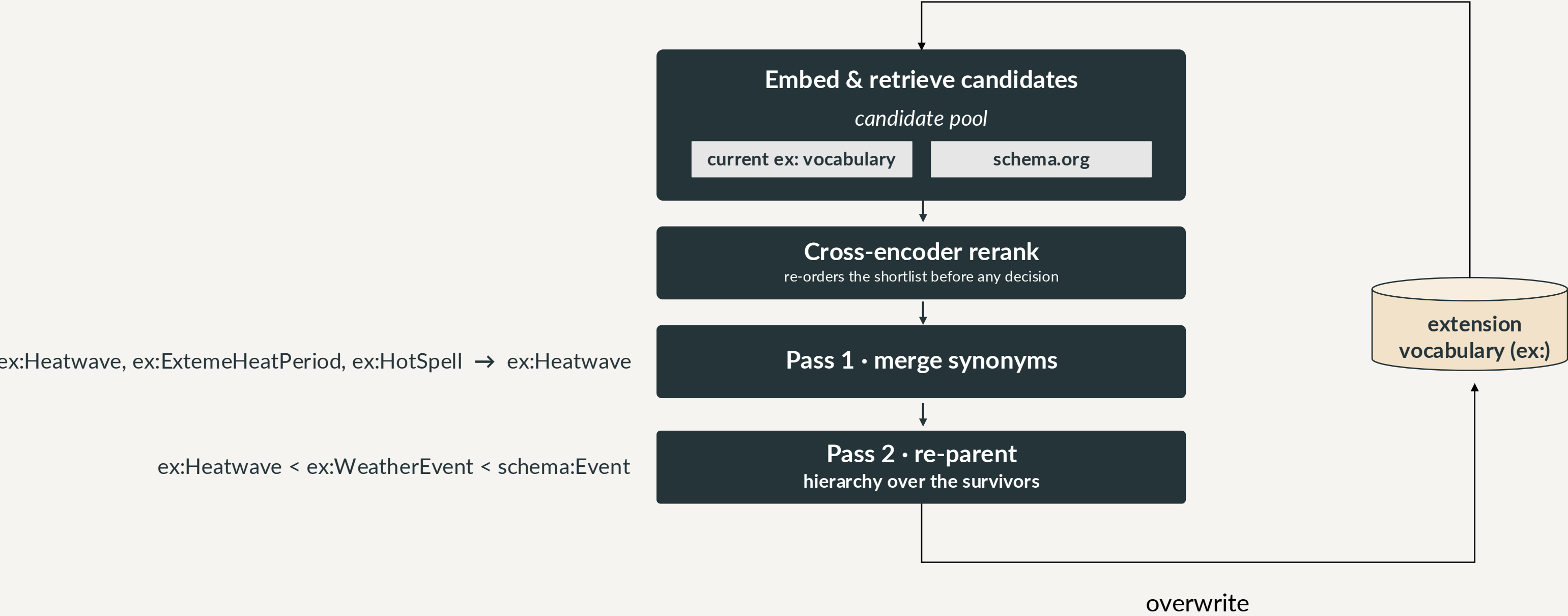
Free-form type	Shortlist (all schema.org)	Score	Decision and stored justification
Country	Country	0.853	reuse schema:Country "The schema.org Country class directly matches the type and its definition, and 'Iraq' and 'US' are clear instances of countries."
	State	0.825	
	Place	0.781	
Building	GovernmentBuilding	0.838	no adequate match; mint ex:Building under schema:Place "schema.org lacks a general Building class. Place is the most appropriate parent, and the new type covers all mentions like homes and businesses."
	House	0.835	
	CivicStructure	0.821	

Stage 6 • Consolidation of Extension Terms



The Canonicalize step of EDC, plus reranking, with the accept-or-mint decision made explicitly rather than by a similarity threshold.

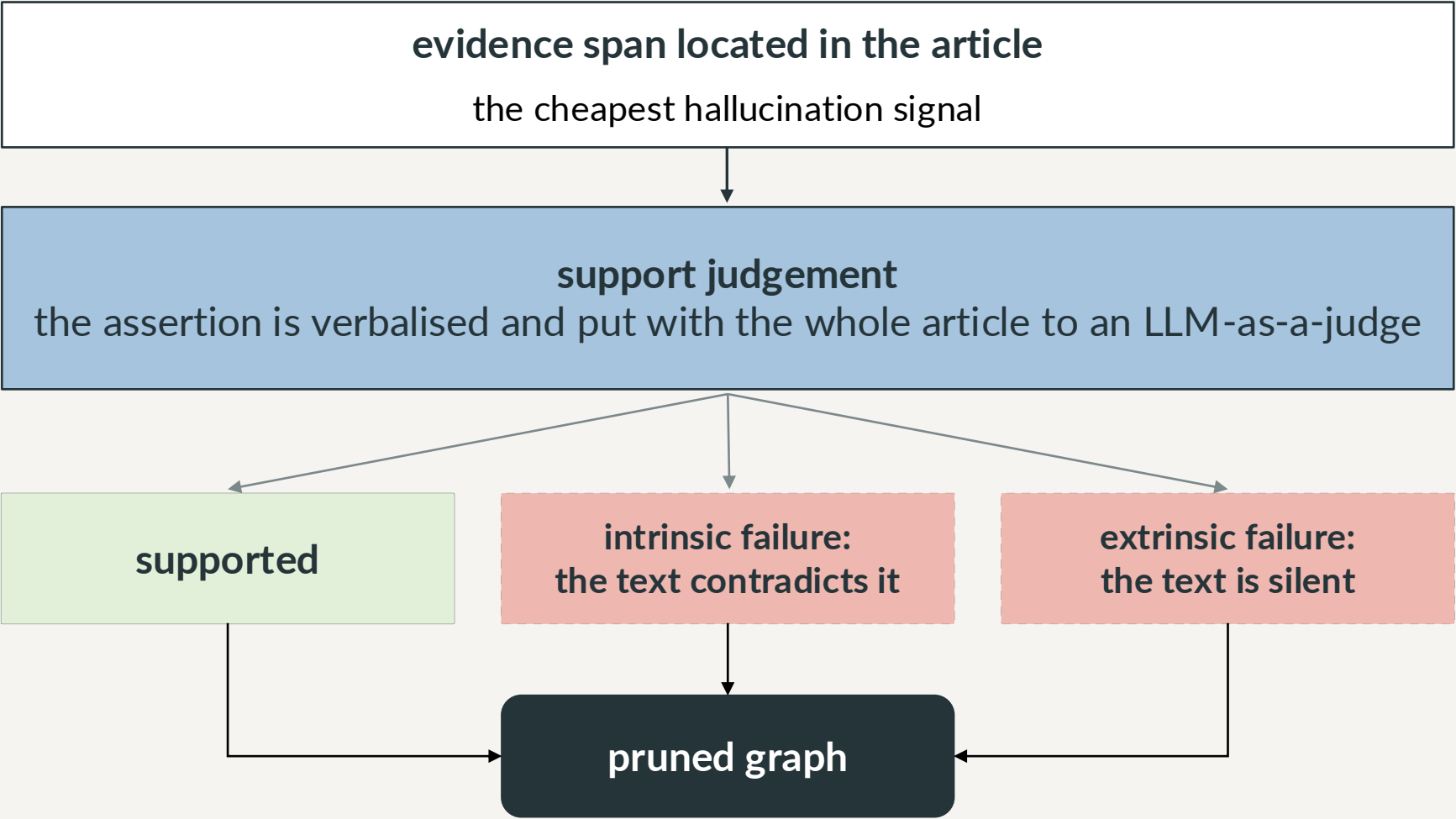
Inside a consolidation pass (runs every 5 articles)



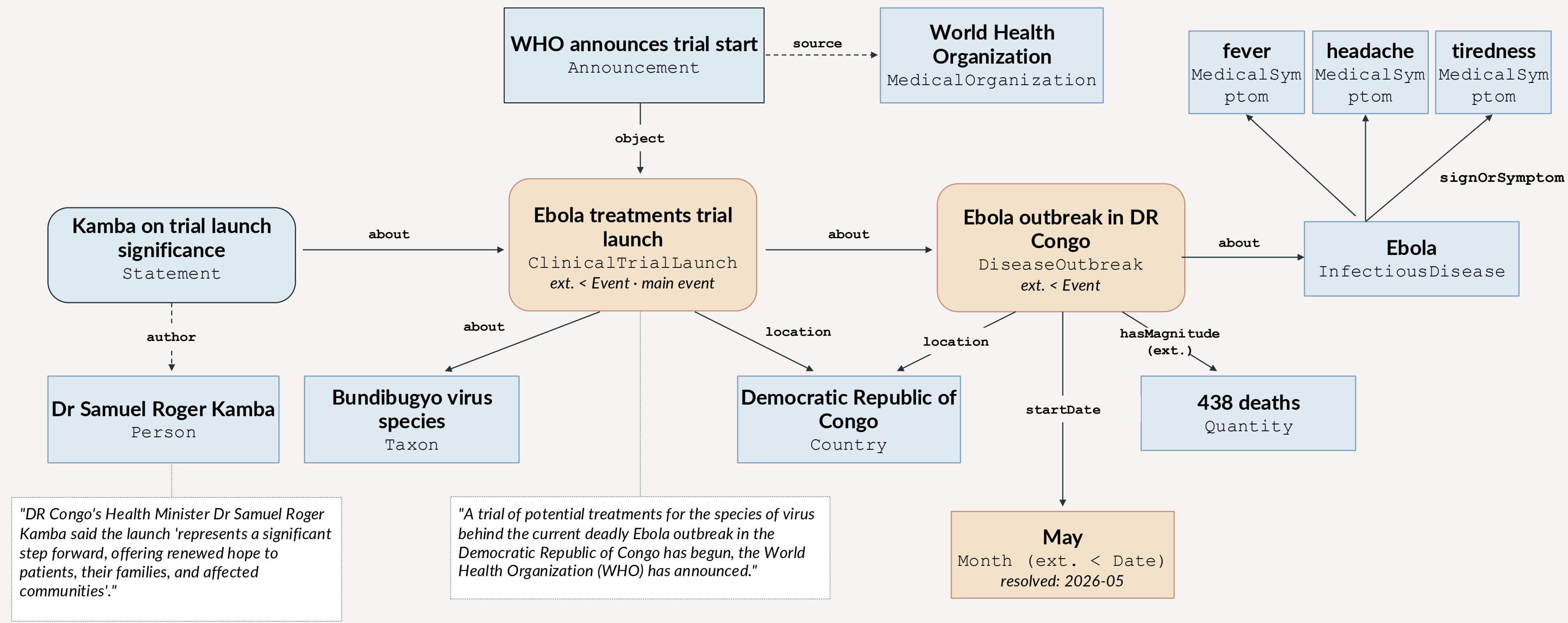
Stage 7 • Grounding



Nothing is deleted: an item that fails keeps its verdict, so every rate in the evaluation can be recomputed from the stored judgements.



What the output looks like: a fragment of a finished graph



Every node and edge also carries its evidence spans, parent and the model's justification (omitted here).

Legend:

- reified event (ext. class, resolves to Event)
- entity on a schema.org class

Evaluation Instruments and their audits



Is Graph true to text?

Grounding Judge Benchmark:

- Used to choose the LLM-as-a-judge (gpt-oss-120b, acc=92.6%)
- 475 hand-labelled assertions, half intact, half deliberately corrupted

Are the article facts recoverable from Graph?

MINE-1 benchmark (adapted)

- **New dataset:** 100 News Articles & 15 hand-written facts per each
- **250-fact audit** to measure accuracy of the LLM-as-a-judge (88%, Llama-3.3-70B)

Is the pipeline choosing the right entity types and relations?

Hand Audit of Canonicalisation

- 1,000 relations (250 per LLM)
- For each, answer: **do the subject and object types fit, does the property make sense, did a better schema.org term exist?**

Testing on Four LLMs: gemini-2.5-flash, DeepSeek-V4-Flash, Llama-3.3-70B, Qwen3-32B

Best-Performing Model Results (Gemini-2.5-flash)



87.4%

of relations confirmed
against the article
(Grounding)

81.2%

Relation types chosen
correctly (judged sound by
hand)

82.3%

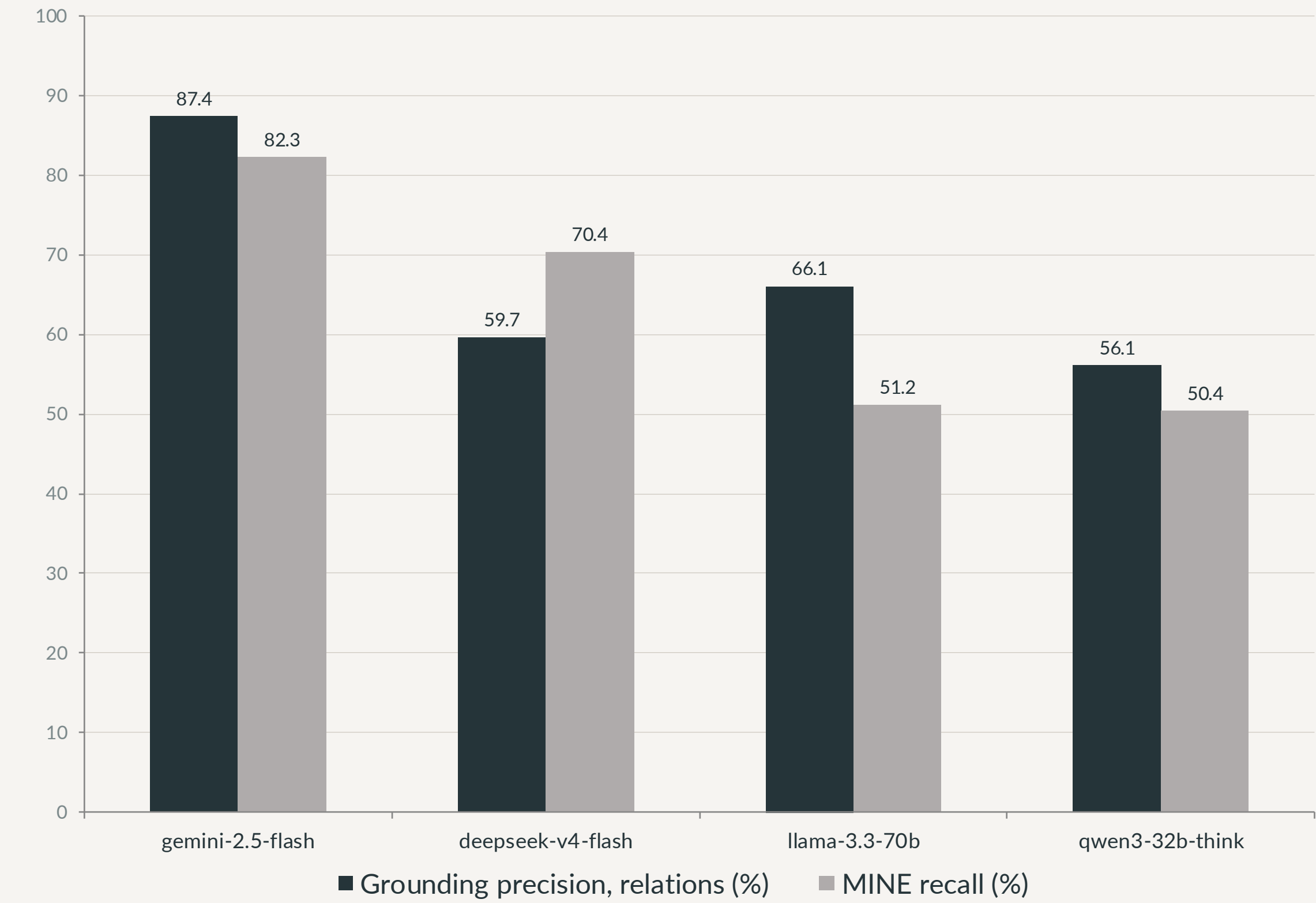
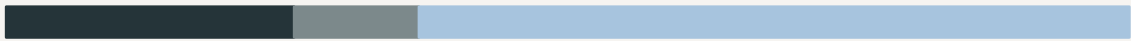
of the reference article
facts recovered from graph
(MINE recall)

64.2%

of triples on standard
schema.org relations

- Judged by **gpt-oss-120b** (grounding) and **Llama-3.3-70B** (recall)
- **Recall** sits within a few points of the whole-graph ceiling; what is missing is absent from the graph.
- Sound = subject type fits, property states a relation that holds between the two arguments in the stated direction, object type fits.

The same pipeline on four LLMs



Volume differs 4 to 7 times

129 relations per graph for gemini against **18** (llama) and **29** (qwen), from an identical prompt and corpus.

Conservatism does not buy precision

The most productive LLM is also the most faithful.

A stronger model lifts every figure without redesign.

Conclusions



01 The gap is real and specific

Publishers annotate correctly but sparsely; the markup describes the page, not what it says.

02 The graphs are largely true to their sources

87.4% of relations and 99.2% of reified nodes confirmed on 80 held-out articles, by a judge whose error was measured first.

03 Most of what an article states is recovered

82.3% of independently written facts; what is missing is absent from the graph, not unretrieved.

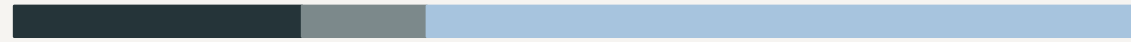
04 Reification helps twice

It represents **n-ary news facts**, and it **results in more use of schema.org terms**.

05 The design is portable; capability varies

Same configuration on a frontier model and on open weights; differences are **of degree**.

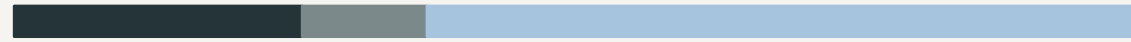
Overall



schema.org is a workable target for extracting what news articles say.

Pipeline has a great potential in providing KGs for richer semantic annotations, considering it is the first combining all the presented ideas together.

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Thank you !

Appendix

Quantifying the Motivation: Analysis of Existing Semantic Annotations



119 news and fact-checking domains from the Web Data Commons release of Common Crawl, analysed whole in DuckDB: 69.6 M quads, 12.2 M entities.

annotations use **valid
schema.org syntax**

5.75%

of a type's properties
are filled, on average

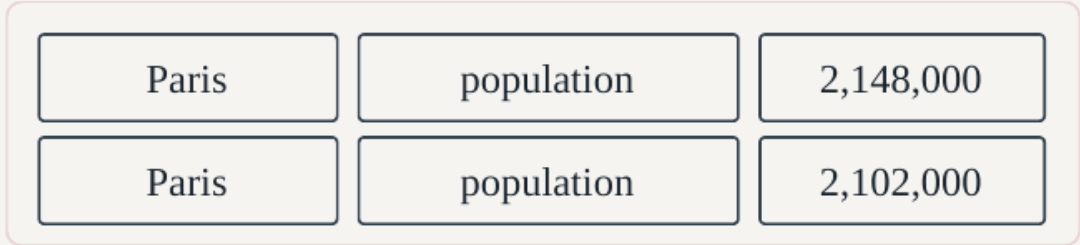
< 0.1%

of article pages fill
'mentions'

**templated
metadata fields**

Background Key Concepts: Reification

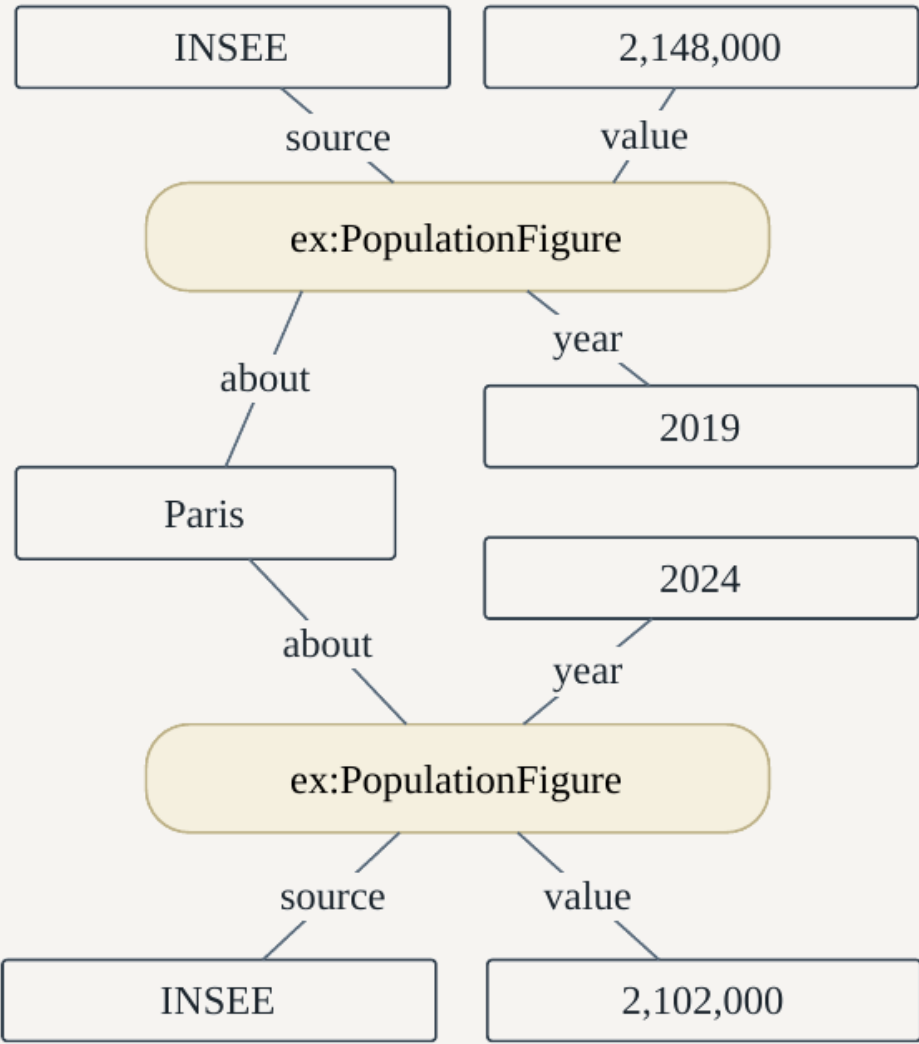
flattened into binary triples



Same subject, same property, two values. As written, the graph contradicts itself.

What separates them is the year each figure was measured in. A year is neither the subject nor the object of the relation, so a binary triple has nowhere to record it.

reified as one node per measurement



Both figures point at the same Paris node, so the city stays a reusable entity, while each measurement keeps the year and the source that tell the two apart. Nothing contradicts.

An event or claim represented as a node of its own, with its participants, values, time, place and source attached.

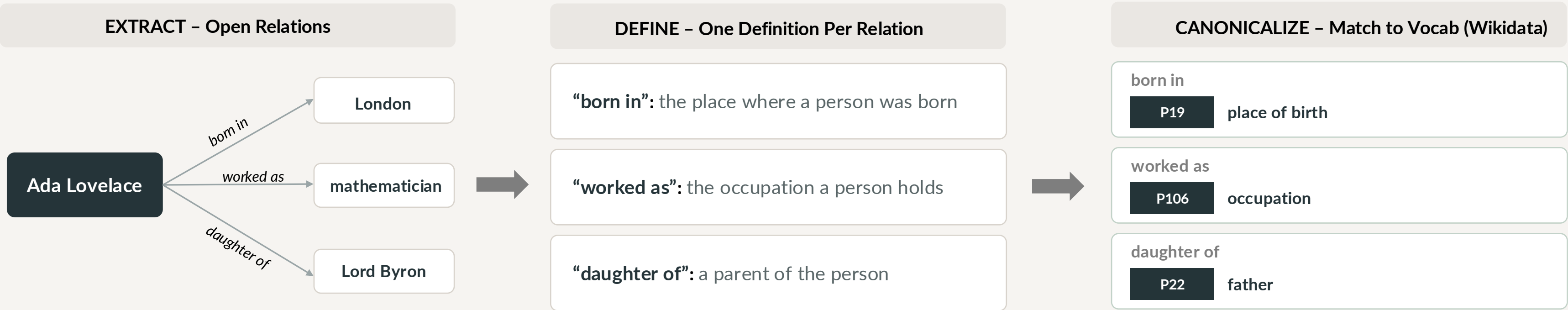
This is the **n-ary relations pattern**, which schema.org supports directly through Event and Claim nodes.

Figure 3.4 of the thesis: two population figures for Paris, flattened (left) and as one node per measurement (right).

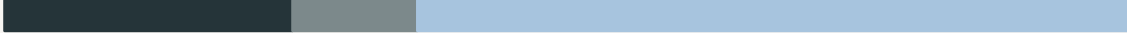
Background Key Concept: The Extract-Define-Canonicalize Method (Zhang & Soh, 2024)



SOURCE TEXT “Ada Lovelace, daughter of Lord Byron, was born in London, and worked as a mathematician.”



Contributions



01 A pipeline that extracts what an article says, in the Web's vocabulary

02 Entity typing and relation typing treated as similarly important

03 A representation built for what news reports

04 Four additions to the extract-define-canonicalise loop

05 A data-driven record of where schema.org falls short for news

06 An evaluation dataset built for the setting

Challenges That Shape The Design



01 The vocabulary is fixed and uneven

Deep for products, **shallow for news**: an Event class, but no heatwave, no airstrike.

02 Types matter as much as relations

The **type of a node governs** what may be said about it. Prior work canonicalises relations only.

03 News facts are not binary

"2,025 excess deaths, 22 to 28 June, a 29% rise" is **one fact with four arguments**.

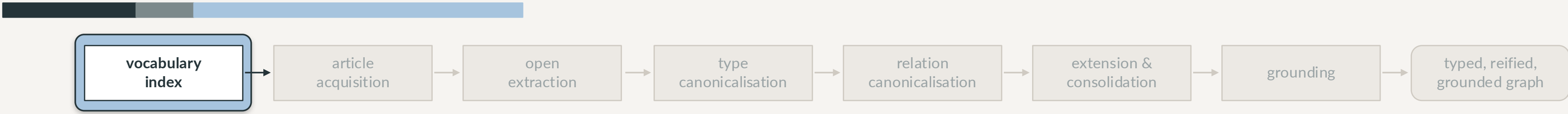
04 Output must be trustworthy

LLMs can hallucinate, so we need to ensure faithfulness.

05 No evaluation dataset

No resource pairs articles with schema.org triples

Stage 1 • Vocabulary Definitions Enrichment



The problem

334 classes and **281 properties** carry a comment that only restates the label.

Fixed template

name: definition [Also called synonyms]

Property glosses are direction-neutral

"an employment relation between a person and an organization", so a property and its **inverse stay in the same shortlist**.

Embedded once, reused by every article

933 classes and **1,521 properties** embedded with **e5-base-v2** and cached.

schema:Museum

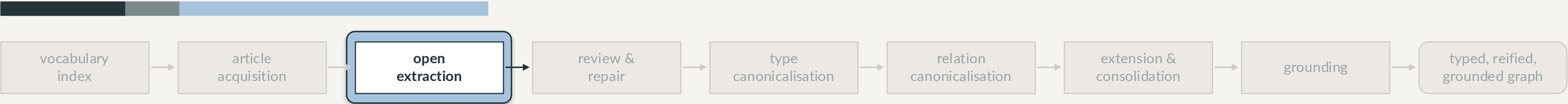
Official comment, in full:

"A museum."

After enrichment:

"A civic structure dedicated to preserving and exhibiting artifacts, art, or scientific specimens. Also called gallery, exhibition hall, cultural institution. Examples: The Louvre. Distinct from Library, which primarily lends books and media."

Events and Claims Nodes



The rules under which reification is applied (Section 4.8); the node, its type and every edge go through the same canonicalisation.

Flattened into binary triples (not adapted here)

```
:France ex:recordedExcessDeaths "2,025" .
:France ex:hadWeeklyIncreaseOf "29%" .
:France schema:affectedBy :heatwave .
```

Context is pushed into the predicate name

Results in more minting

Reified as one n-ary fact node (Event)

```
:m1 a ex:ExcessMortality ;
    schema:location :France ;
    ex:deathCount "2,025" ;
    schema:temporalCoverage "2026-06-22/2026-06-28" ;
    ex:rateOfChange "29%" ;
    ex:cause :heatwave .
```

Edges become short, general and reusable.

Scope

Only **genuine events and attributed claims** are asked to be reified;

stable relationships stay plain edges.

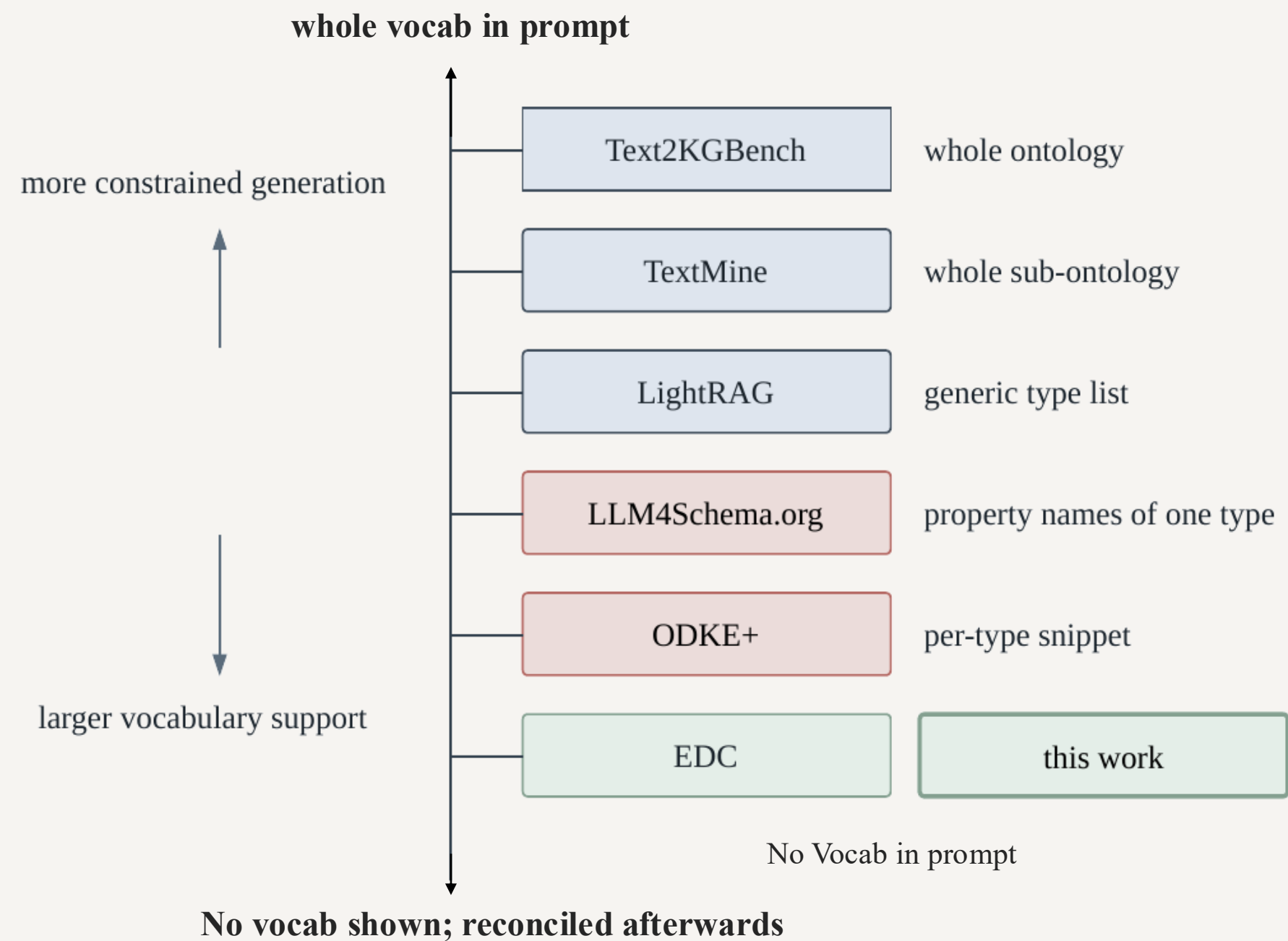
Ordinary JSON-LD

A typed node with properties, so it **serialises as schema.org markup** unchanged.

Typed like any entity

ex:ExcessMortality and ex:heatwave are canonicalised as a type and **resolve upward to schema:Event**.

Related work: where the target vocabulary reaches the model



Vocabulary in the prompt

- Maximal guidance and **high conformance**
- Affordable only for **small ontologies** (Text2KGBench: 20 classes, 68 relations).

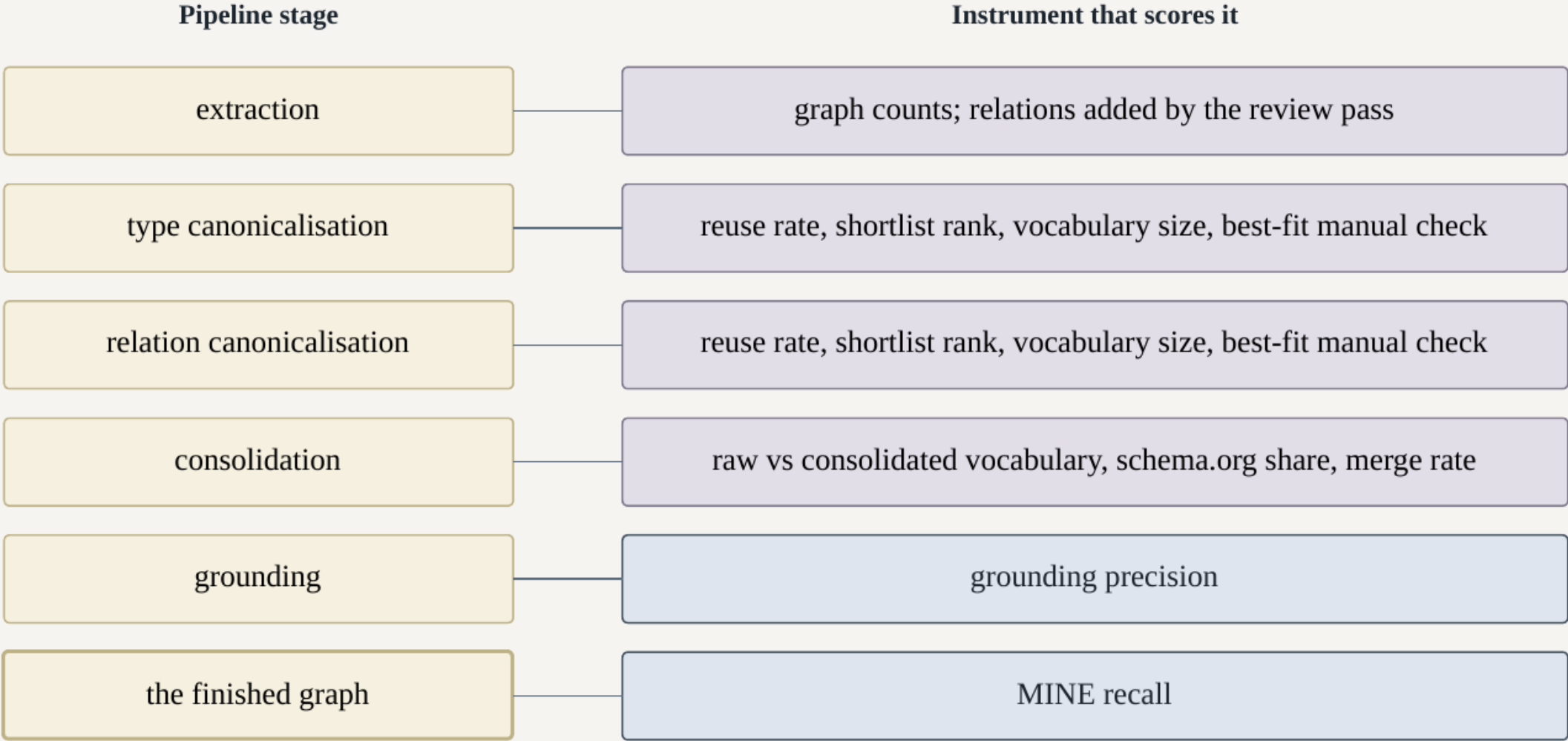
Nothing in the prompt

The best option for our **target vocab.**

schema.org is small enough to sit in a prompt, however:

- **Model falls back on the types already common on Web.**
- **Model finds fewer entities and relations than open extraction.**

Evaluation design



Four LLMs:

gemini-2.5-flash
Llama-3.3-70B
Qwen3-32B
DeepSeek-V4-Flash

The Grounding Judge Benchmark: 475 hand-labelled assertions, half corrupted

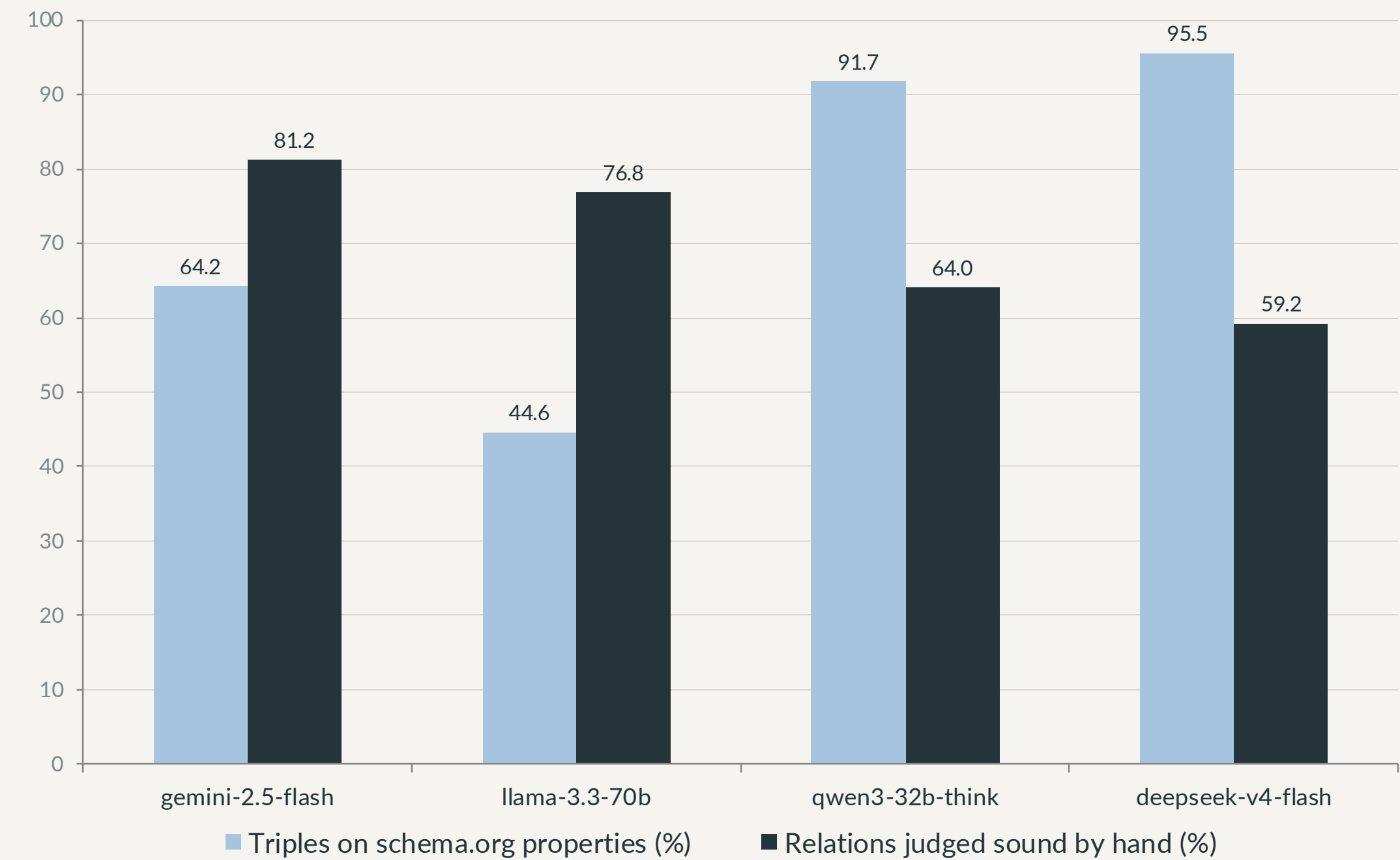
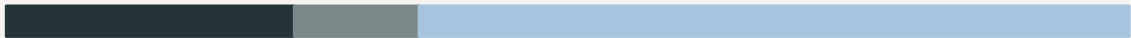


Judge	Recall on corrupted (%)	False-positive rate (%)	Accuracy (%)
gpt-oss-120b	91.3	6.0	92.6
qwen3-32b-think	85.7	2.8	91.2
deepseek-v4-flash	83.5	4.1	89.5
llama-3.3-70b	81.4	3.5	88.6

Three runs each; ranges in brackets. All judges are open-weight with pinned builds, so every figure is reproducible without a vendor.

Which judge	The shared blind spot
gpt-oss-120b catches 91.3% of corrupted assertions, 6.0% false-positive rate makes grounding a few points conservative.	reversed relation direction (69.3% even for the best).

Schema.org Reuse vs correctness



Real properties, wrong sense

- France schema:lifeEvent Heatwave
- France schema:owns St phanie Rist

What it means

- The LLM with the highest reuse rate has the lowest soundness
- Faithfulness must be reported beside conformance

Sound = subject type fits, property states a relation that holds between the two arguments in the stated direction, object type fits.

The effect of each design choice



Reification

+80%

relations expressed, same articles and backend

35% → 69%

triples on a schema.org property

Cross-encoder reranker

48-51%

of the shortlist the model sees was below the bi-encoder cutoff

19.5%

of gemini decisions chose a promoted standard property

On validation: schema.org share **50.4 → 70.2%**, vocabulary **123 → 76 terms**.

Review pass

+5%

net relations on gemini

+98%

net relations on llama-3.3-70b

Worth most **where the first call left most**.

Consolidation

55%

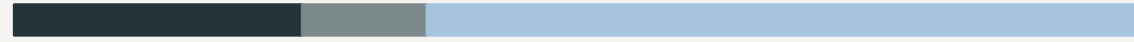
Reduction in extension size on llama and qwen; -7% on gemini

+19 / -13

points reuse up, grounding down on deepseek

Folds mints into standard properties, but at scale **can over-merge**.

The naive baseline



The whole vocabulary in one prompt, same validation articles:

Model falls back on the types already common on web

No event or claim nodes

18% of relations name an undefined identifier

A Third of all relations collapse onto 'about' or 'location'

Some articles truncated (due to long context)

Graphs are 1/3 the size in relations

Next steps, based on limitations



- **Solve scalability issue by incremental consolidation:** judge only new mints, retrieve against everything. restores linear cost
- **A directional check** using the domain-to-range signature the vocabulary already carries (weakest grounding errors are just wrong relation direction)
- **Experiment with stronger models:** the most capable backend led on every axis
- **A larger corpus**, including Wikipedia-style sources that need chunking